Angled and negative work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use force direction to determine work's sign and size.

SUCCESS CRITERIA - I can...

1. Compare force direction with displacement.
2. Use the parallel force component.
3. Explain positive, negative and zero work.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A positive triangular pulse peaks at 50 N and lasts 0.3 s. It acts on a 3 kg cart initially at -2 m/s. Find impulse and final velocity.

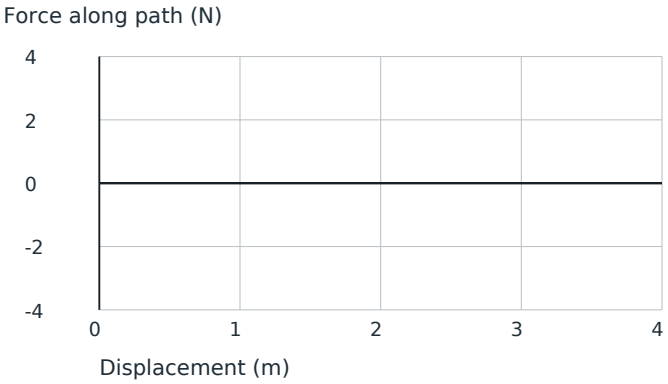
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $W = Fd\cos\theta = F_{parallel}\, d$.
θ is between force and displacement. CPA uses the supplied parallel component. Opposite force gives negative work.

Worked reasoning: A box moves 4 m right while friction acts 3 N left. Friction's work is -3×4=-12 J because force opposes displacement.

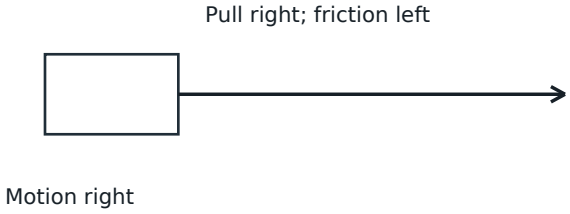
Your turn: Draw force and displacement arrows. Add a perpendicular upward force and state its work for this horizontal motion.



Marzano check - circle: ① ② ③

03 Read and interpret the model

The box moves right. Classify work by a rightward pull, leftward friction and upward normal force.



The rectangle is the box. Add the requested force and displacement labels.

Marzano check - circle: ① ② ③

04 Apply the physics

A 20 N rope pulls a crate at 30° above horizontal as the crate moves 5 m horizontally. Friction is 4 N. Draw and label the force angle and displacement. Calculate rope work, friction work and net work.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A force perpendicular to velocity can bend a path without changing speed. Explain using work.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Fragments of masses 4 and 6 kg separate from rest. The 4 kg piece moves at +12 m/s. Find the other velocity and total kinetic energy.
Use before and after snapshots to organize the momentum account.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A 10 N force acts at 120° to a 2 m displacement. Find work and explain the sign.

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 20 N pull at 60° above horizontal moves a box 3 m right while friction is 4 N. Find work by pull, friction and their sum.

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3